

AlphaDynamics: Coupled Phase Oscillators for Protein Torsion Dynamics Prediction

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Abstract

We introduce AlphaDynamics, a compact per-system neural propagator for protein conformational dynamics operating directly on the backbone torsion manifold $\mathbb{T}^N = (\mathbb{S}^1)^{2N}$. The model couples N phase oscillators via learnable pairwise interactions and evolves them under a continuous ordinary differential equation solved by a fourth-order Runge-Kutta adjoint integrator. Final phases are mapped to a mixture of axis-independent von Mises densities on $\mathbb{T}^N$ through a shallow feed-forward head. Unlike transferable surrogates such as Timewarp (Klein et al. 2023), which amortise a single large model across many peptides, AlphaDynamics trains a lightweight specialist (348K parameters) per protein domain. On a 57-domain subset of the mdCATH benchmark (Mirarchi et al. 2024), covering two size regimes (37 domains at $N_{\text{res}} = 50$ and 20 domains at $N_{\text{res}} = 100$) under an identical simulation protocol (CHARMM36m + TIP3P water at 348 K, five replicas per domain), AlphaDynamics outperforms a matched multilayer-perceptron baseline on every domain, with a mean NLL advantage of $6.4\times$ at $N_{\text{res}} = 50$ growing to $7.5\times$ at $N_{\text{res}} = 100$. A 2500-step autoregressive rollout test shows that predicted trajectories remain stable without exponential blow-up, though rollout fidelity requires a concentration-rescaling heuristic ($\kappa \rightarrow 30\kappa$) whose replacement remains an open problem. Ablation experiments confirm that the ODE integration is the critical architectural component. We report two empirical observations: (i) the optimal ODE integration horizon t_{max} correlates with the temporal correlation length of the training data, and (ii) the NLL advantage over the MLP baseline is largest for conformationally ordered domains.

1. Introduction

The static protein-structure prediction problem, long considered one of the grand challenges of computational biology, has been substantially addressed by AlphaFold 2 and its successors (Jumper et al. 2021; Abramson et al. 2024). Attention has consequently shifted toward the harder question of conformational dynamics: how a folded protein moves in time, samples alternative states, and responds to ligands or mutations. Classical molecular

dynamics (MD) (Hollingsworth & Dror 2018) provides an exact but expensive answer; current state-of-the-art simulations rarely exceed millisecond time scales and require specialized hardware (Lindorff-Larsen et al. 2011).

Three families of machine-learning surrogates have emerged. First, equilibrium samplers such as bioEmu (Lewis et al. 2024) and AlphaFlow / ESMFlow (Jing et al. 2024a) generate statistically independent structural ensembles conditioned on sequence, typically by combining an AlphaFold-style structure predictor with flow-matching or score-based generative modelling. These methods excel at enumerating metastable states but do not return trajectories and therefore cannot quantify kinetic observables such as transition rates. Second, AlphaFold-augmentation methods subsample the multiple-sequence alignment or introduce experimental restraints (NMR, DEER) to coax alternative conformations from AlphaFold 2 (Wayment-Steele et al. 2024; Sala et al. 2024). These preserve the static-prediction paradigm and again lack temporal continuity. Third, and most closely related to our work, neural propagators such as Timewarp (Klein et al. 2023) and MDGen (Jing et al. 2024b) predict the conformation of a peptide at a future time conditioned on its current Cartesian coordinates and velocities. Existing propagators operate in full Cartesian space and rely on large normalizing-flow or transformer-based architectures (Timewarp contains 396 M parameters), which limits deployability.

In this work we take a different route. We observe that backbone dynamics lives on a product of circles, $\mathbb{T}^N$, and that the natural primitives for such a manifold are not Cartesian vectors but phase oscillators. Coupled phase oscillator networks have long been studied in non-linear dynamics (Kuramoto 1984; Strogatz 2000) and recently re-emerged as a general computational substrate (Muscinelli et al. 2024; Gwózdź 2026a). We combine a learnable pairwise coupling between oscillators, evolved as a continuous ODE, with a mixture-of-von-Mises output head that respects the underlying torus topology. Ablation experiments (§4.9) confirm that the ODE integration is the critical component. The entire system is trained end-to-end by back-propagating through a Runge-Kutta adjoint ODE solver (Chen et al. 2018).

We emphasise that AlphaDynamics is a per-system surrogate: a separate lightweight model is trained for each protein domain of interest. This contrasts with transferable surrogates such as Timewarp (Klein et al. 2023), which train a single large model on a library of peptides and generalise to unseen sequences. The two paradigms serve different use cases. Transferable surrogates are preferable when one needs rapid predictions across many systems without retraining; per-system surrogates are preferable when a single protein is studied in depth and training cost is low (AlphaDynamics trains in under 10 minutes on a single GPU). Direct parameter-count comparisons between the two paradigms are misleading, as transferable models amortise their capacity across many systems.

The contributions of this paper are:

1. **Architecture.** We present AlphaDynamics, a phase-oscillator neural propagator for protein torsion dynamics with 348K trainable parameters per domain. The architecture exploits the torus topology of dihedral angles through coupled oscillator dynamics rather than Cartesian coordinates (§3). We note that this per-system design differs fundamentally from transferable surrogates like Timewarp (396M parameters amortised across peptide families); the two approaches address complementary use cases.
2. **Uniform benchmark.** We evaluate on a 57-domain subset of the mdCATH dataset (Mirarchi et al. 2024) under an identical simulation protocol across domains and two size regimes ($N_{\text{res}} \in \{50, 100\}$), comparing against a matched MLP baseline, a per-residue identity predictor, and a GAT-GNN baseline (§4).
3. **Empirical observations.** We report two trends: the optimal ODE integration horizon correlates with the temporal correlation length of the training data, and the NLL advantage over the MLP baseline is largest for conformationally ordered domains (§4.4). These observations are preliminary and require validation on larger and more diverse protein sets.
4. **Rollout stability.** Long (2500-step) autoregressive rollouts do not diverge, though distributional fidelity requires a concentration-rescaling heuristic ($\kappa \rightarrow 30\kappa$) at sampling time. We report this honestly as a current limitation (§4.5).
5. **Inference cost.** On a single GPU, AlphaDynamics generates one nanosecond of predicted trajectory in approximately 16 ms per domain. We compare this to the wall-clock cost of explicit MD on the same domain rather than to transferable surrogates operating on different tasks (§4.10).

2. Background

2.1 Torsion-space molecular dynamics

The backbone conformation of a protein with N_{res} residues is fully described, up to ideal bond lengths and angles, by the pairs of dihedral angles $(\varphi_i, \psi_i) \in [-\pi, \pi]^2$ for $i \in \{1, \dots, N_{\text{res}}\}$. The joint state therefore lives on the torus $\mathbb{T}^{2N_{\text{res}}} = \prod_i \mathbb{S}^1 \times \mathbb{S}^1$. Representing conformations directly on this manifold avoids the topological artefacts of mapping onto $\mathbb{R}^n$ or $\mathbb{S}^2$.

2.2 Coupled phase oscillators and phase-gate coupling

A Kuramoto network (Kuramoto 1984) of M phase oscillators with natural frequencies ω_i and pairwise couplings K_{ij} evolves as $\dot{\phi}_i = \omega_i + \sum_j K_{ij} \sin(\phi_j - \phi_i)$. Such networks exhibit rich collective behaviour (synchronization, chimera states).

In prior work (Gwózdź 2026a), we showed that a modified coupling of the form $K \cos(\phi_c) \sin(\phi_t - \phi_{out})$ implements a classical CNOT (controlled-NOT) logic gate via pure oscillator dynamics: when the control phase $\phi_c \approx 0$, $\cos(\phi_c) \approx +1$ and the output synchronises with the target; when $\phi_c \approx \pi$, $\cos(\phi_c) \approx -1$ and the output anti-synchronises (flips the target bit). This sign-modulated injection locking achieves 100% gate accuracy under noise amplitudes up to $a = 1.0$ across 20 random seeds, and together with NOT and AND gates (also implemented as oscillator ODEs) yields a functionally complete gate set. With bistable D-latch memory, the system is Turing-complete under standard unbounded-memory assumptions (Gwózdź 2026a, §4 and Formal Appendix).

In AlphaDynamics we generalise this coupling to a learnable pairwise interaction: $W_{ij} \cos(\phi_j) \sin(\phi_j - \phi_i)$, where W is an asymmetric $M \times M$ weight matrix. Each entry W_{ij} can be interpreted as the strength of a gated interaction from oscillator j to oscillator i —the same sign-modulation mechanism as the CNOT gate, but with learned coupling strengths adapted to the dynamical structure of the protein. The $\cos(\phi_j)$ factor acts as a gain modulator: oscillator j exerts maximal influence when its phase is near 0 or π (the two logical states) and minimal influence at $\pm \pi/2$.

2.3 Mixture of von Mises densities

A von Mises distribution on S^1 has density $p(\phi; \mu, \kappa) = \exp(\kappa \cos(\phi - \mu)) / (2\pi I_0(\kappa))$, where μ is the circular mean and $\kappa \geq 0$ the concentration.

Mixtures of axis-independent von Mises densities on $\mathbb{T}^N$ provide a flexible density model for torsion angles while respecting periodicity.

3. Method

The full pipeline is summarized in Figure 5.

3.1 Phase-flow encoder

Given an input conformation $(\boldsymbol{\varphi}, \boldsymbol{\psi}) \in \mathbb{T}^{2N_{\text{res}}}$ we lift to M oscillator phases $\{\theta_k\}_{k=1}^M$ by an affine map

$$\theta_k^{(0)} = w_k^\varphi \cdot \boldsymbol{\varphi} + w_k^\psi \cdot \boldsymbol{\psi} + b_k.$$

The phases then evolve under the ordinary differential equation

$$\frac{d\theta_k}{dt} = \omega_k + \sum_{j=1}^M W_{kj} \frac{\cos(\theta_j - \theta_k) + a \sin(\alpha_k - \theta_k)}{2},$$

integrated from $t = 0$ to $t = t_{\text{max}}$ with a fourth-order Runge-Kutta adjoint scheme (Chen et al. 2018). The natural frequencies ω_k and anchors α_k are initialized to break symmetry and avoid resonances; ablation (§4.9) shows these choices have negligible impact compared to the ODE integration itself. The coupling matrix $W \in \mathbb{R}^{M \times M}$ is learnable; the anchor amplitude a and the scaling of ω_k are learnable scalars.

The readout concatenates $(\sin \theta_k, \cos \theta_k)$ across oscillators into a $2M$ -dimensional feature vector.

3.2 Mixture-of-von-Mises head

A shallow multilayer perceptron (two hidden GELU layers of width 128) maps the oscillator features to the parameters of a K -component mixture of axis-independent von Mises densities on $\mathbb{T}^{2N_{\text{res}}}$: mixture weights $\pi_1 : \pi_K$, circular means $\mu_{k,i}$, and concentrations $\kappa_{k,i} > 0$ for every component k and every torsion angle i . We use $K = 8$.

3.3 Training

Given pairs of consecutive frames $(\mathbf{x}_t, \mathbf{x}_{t+\Delta t})$ drawn uniformly from a trajectory, we minimize the negative log-likelihood $-\log p(\mathbf{x}_{t+\Delta t} | \mathbf{x}_t)$ of the target under the predicted mixture. We use the AdamW optimizer (Loshchilov & Hutter 2019) with learning rate 2×10^{-3} , weight decay 10^{-4} , cosine schedule, and 4000 gradient steps with batch size 512. Each domain is trained independently (no multi-domain fine-tuning).

3.4 Baseline

We compare against an MLP baseline that takes the same $(\sin \phi_i, \cos \phi_i, \sin \psi_i, \cos \psi_i)$ input, passes it through three GELU layers of width 128, and produces the parameters of an identical mixture head. The baseline uses slightly more parameters ($\sim 396\text{K}$) than AlphaDynamics ($\sim 348\text{K}$) so the comparison is conservative.

3.5 GNN baseline

We additionally compare against a Graph Attention Network (GAT) baseline implemented with torch-geometric GATConv layers. The graph is a linear chain (residue i connects to $i \pm 1$), with 3 GAT layers, 4 attention heads, node dimension 64, and the same MDN output head as AlphaDynamics. The GAT baseline contains 875K parameters—more than twice AlphaDynamics (342K)—giving it a capacity advantage.

4. Experiments

4.1 Dataset

We use mdCATH (Mirarchi et al. 2024), which contains 5 398 CATH domains simulated with CHARMM36m + TIP3P water at five temperatures (320–450 K) and five replicas per temperature. For the main benchmark we select all 37 domains with exactly 50 residues, and for the scaling study 20 randomly chosen domains with exactly 100 residues. We always use the 348 K temperature (single-replica aggregation of all five replicas for statistics). Trajectories are converted to ϕ, ψ angles via mdtraj (McGibbon et al. 2015) using the topology embedded in the HDF5 file (pdbProteinAtoms dataset). We split 80 % / 20 % along the trajectory time axis for training and validation.

4.2 Main benchmark results

Table 1. Summary statistics across the two size regimes. “AD wins vs MLP” counts domains where AlphaDynamics NLL < MLP NLL. “Mean identity” is the mean per-frame angular change in degrees—the error of a trivial predictor that copies the current frame as its prediction.

Size (N_{res})	Domains	Mean identity ($^\circ$)	Mean MLP NLL	Mean AD NLL	AD wins vs MLP	Mean ratio
50	37	34.7	580.0	108.0	37/37	6.4×
100	20	26.3	686.9	109.3	20/20	7.5×
Combined	57	—	—	—	57/57	—

We include the identity baseline (predicting zero change) to contextualise the difficulty of each domain: lower identity values indicate more ordered proteins with smaller per-step conformational changes. Note that the identity predictor and the probabilistic models (MLP, AlphaDynamics) are not directly NLL-comparable because the identity predictor is deterministic; the identity column serves as a descriptor of domain difficulty, not as a competing model.

Figure 1 visualizes the per-domain NLL pairs: every point lies below the parity diagonal. Figure 3 shows that MLP performance degrades substantially from $N_{\text{res}} = 50$ to $N_{\text{res}} = 100$ (mean NLL $580 \rightarrow 687$), whereas AlphaDynamics is nearly flat ($108 \rightarrow 109$), indicating that phase coupling scales more efficiently with chain length than feed-forward processing of joint torsion features.

4.3 Cross-temperature generalization

To test whether AlphaDynamics generalizes beyond training temperature, we trained on 348 K only and evaluated on all five mdCATH temperatures (320, 348, 379, 413, 450 K) across 5 representative domains.

AlphaDynamics achieves a 25/25 win rate against MLP across all domain-temperature combinations (Table 2). Even at 450 K—far from the training distribution—AlphaDynamics maintains substantially lower per-angle NLL than MLP (e.g. 0.70 vs 3.29 on 1kwgA03).

4.4 Empirical observations

Observation 1 — warmup scaling (pilot, $n=4$ domains). We swept $t_{\text{max}} \in \{1, 2, 4, 8\}$ on four pilot 50-residue domains and found $t_{\text{max}} = 4$ to be optimal in every case. Exploratory runs on longer-time-gap data (e.g. 19.2 ns jumps from the Timewarp tetrapeptide dataset (Klein et al. 2023)) showed that the optimum shifts to smaller t_{max} when the data are drawn from enhanced-sampling protocols with shorter effective temporal correlations. We stress that this observation is based on a small pilot scan; a systematic sweep across all 57 benchmark domains is left to future work.

Observation 2 — order-dependent advantage. The ratio $\text{NLL}_{\text{MLP}} / \text{NLL}_{\text{AlphaDynamics}}$ is inversely correlated with the identity baseline of the domain (the per-frame conformational change): on the most ordered 50-residue domain (1lwjA03, identity = 25.8°) the ratio is $11.2\times$, while on the most disordered (1vq8L01, identity = 55.0°) it is $3.6\times$. Phase coupling is more beneficial when dynamics have learnable temporal structure. Figure 2 shows the log-linear relationship across both size classes.

4.5 Long-rollout stability

We selected five 50-residue domains spanning the full identity range and ran a 2500-step autoregressive rollout with concentration re-scaling $\kappa \rightarrow 30\kappa$ at

sampling time (chosen empirically; see §5 for discussion). Step magnitudes decrease slightly over the rollout (mean drift -32° joint, i.e. no exponential divergence). Per-residue Ramachandran Kullback-Leibler divergence against the full reference distribution averages 1.84, compared to a ground-truth-vs-ground-truth KL of ~ 0.1 . The rollout distribution is qualitatively correct but narrower than the reference—consistent with the over-concentration caveat in §5. Figure 4 reports both metrics per domain.

4.6 Ramachandran free-energy fidelity

We computed per-residue free-energy surfaces $G(\phi, \psi) = -RT \ln P$ from 2500-step rollouts and compared to ground truth via four metrics: Jensen-Shannon divergence (JSD), marginal Wasserstein distance (EMD), basin-center $|\Delta G|$, and basin population error. On the two most ordered domains (1lwjA03, 1kwgA03), JSD averages 0.148 (good; <0.1 is excellent) and $|\Delta G|$ averages 1.0 kcal/mol (thermal $k_B T$ scale). The most disordered domain (1vq8L01) shows higher JSD (0.335) due to the κ -rescaling artifact discussed in §5. Figure 7 shows representative Ramachandran free-energy maps.

4.7 GNN baseline comparison

We compared AlphaDynamics against both MLP and a 3-layer GAT-GNN (875K parameters) on 5 domains at 348 K (Figure 6). AlphaDynamics wins 5/5 against both baselines. Notably, GAT-GNN performs worse than MLP on 4/5 domains despite having $2.3\times$ more parameters. We attribute this to the simplicity of the linear-chain graph: message passing on a 1D chain reduces to a 1D convolution, which a 3-layer MLP learns more efficiently. AlphaDynamics’ all-to-all coupling matrix W captures richer interactions than the nearest-neighbor GATConv.

Domain	MLP NLL	GAT NLL	AD NLL	AD/MLP	AD/GAT
1lwjA03	253.8	274.2	31.8	$8.0\times$	$8.6\times$
1kwgA03	139.4	379.1	21.5	$6.5\times$	$17.6\times$
1ss3A00	202.6	569.3	66.4	$3.1\times$	$8.6\times$
1vq8L01	609.2	197.7	139.5	$4.4\times$	$1.4\times$
1hw7A02	544.4	1040.6	105.6	$5.2\times$	$9.9\times$

4.8 Ablation: axis-independent vs correlated MDN head

We tested whether modeling ϕ - ψ correlations explicitly within each mixture component (bivariate von Mises with cross-concentration κ_c ; Singh et al. 2002) improves rollout fidelity. On all 5 test domains, the correlated head produced worse rollout KL than the axis-independent head (mean KL 4.18 vs 2.04). We conclude that the 8-component mixture implicitly captures ϕ - ψ correlations through component placement, and explicit cross-terms add noise without benefit at this training scale.

4.9 Component ablation

To identify which elements contribute most, we tested five variants on 5 domains: the full model, random frequencies (instead of structured), no anchor term, standard Kuramoto coupling (without asymmetric gating), and no ODE (MLP on lifted phases). ODE integration is the dominant contributor: removing it degrades NLL by 654 nats on average ($10\times$ worse). The remaining components (frequency initialization, anchor, coupling form) have negligible average impact ($|\Delta\text{NLL}| < 1.5$), confirming that the model’s strength lies in the continuous dynamical system prior on the torus, not in any particular parameterization of the ODE right-hand side (Figure 8).

4.10 Computational cost

On a single NVIDIA RTX-5090, AlphaDynamics generates one predicted frame (a one-nanosecond step) in ≈ 16 ms including the adjoint integration. For reference, vacuum OpenMM simulation of alanine dipeptide takes ≈ 340 s per nanosecond on the same hardware. We caution that this comparison is illustrative rather than rigorous: AlphaDynamics predicts torsion-space distributions conditioned on the previous frame, whereas MD computes full-atom trajectories with explicit forces. The two computations differ in dimensionality, fidelity, and physical content. Per-domain training time is under 10 minutes.

5. Discussion

Positioning. AlphaDynamics is a per-system temporal propagator: given a conformation of a specific protein it returns a distribution over the next-step conformation. It shares the propagator paradigm with Timewarp (Klein et al. 2023) and MDGen (Jing et al. 2024b), but differs in two fundamental respects. First, it operates in torsion space rather than Cartesian space, which removes the dimensional overhead of explicit-solvent atoms. Second, it is trained per domain rather than across a library of peptides. This per-system design limits applicability to settings where the target protein is known in advance and a short training run is acceptable, but it enables extreme parameter efficiency (348K per domain). Transferable surrogates like Timewarp address the complementary case where rapid zero-shot prediction across many peptides is needed. Equilibrium samplers such as bioEmu (Lewis et al. 2024) and AlphaFlow (Jing et al. 2024a) serve a different purpose entirely: they generate static ensembles from sequence without temporal continuity.

Why coupled oscillators? The coupled-oscillator prior is motivated by two observations. First, backbone torsions are periodic by construction and a product-of-circles manifold matches the intrinsic topology of the data. Second, the learnable coupling matrix W naturally encodes pairwise cooperativity—two residues whose ϕ, ψ angles are mechanically coupled through the peptide bond behave collectively. Empirically, the MLP baseline degrades rapidly as N_{res} grows (NLL $580 \rightarrow 687$ from $N = 50$ to $N = 100$),

whereas AlphaDynamics is nearly flat ($108 \rightarrow 109$). The peptide-bond inductive bias pays dividends at chain scale.

Limitations. Several caveats warrant explicit mention.

Rollout fidelity. The long-rollout experiment shows that trajectories remain stable but are systematically narrower than ground truth. The present inference procedure re-scales the predicted concentrations by $\kappa \rightarrow 30\kappa$ to reduce per-step drift, and this value is not optimal for every domain. We tested multi-step training with $K=4$ rollout steps and straight-through von Mises sampling as an exposure-bias mitigation; on all 5 test domains this degraded rollout fidelity compared to simple κ -rescaling (mean KL 4.80 vs 2.04). The κ -rescaling heuristic therefore remains the recommended inference procedure.

Cross-temperature. We verified cross-temperature generalization (§4.3) with a 25/25 win rate across 320–450 K, though all training uses 348 K only.

Head-to-head comparison. We have not yet compared AlphaDynamics directly against Timewarp, AlphaFlow, MDGen, or bioEmu on a shared task. Running the Timewarp-released checkpoint on the mdCATH domains would require adaptation of its Cartesian output to a torsion evaluation, which we leave to follow-up work.

Metric choice. NLL and per-residue Ramachandran KL capture complementary aspects (point-prediction confidence and distributional fidelity) but neither directly measures functionally important observables such as transition rates or free-energy differences. CASP-style refinement targets (Kryshtafovych et al. 2023) or D.E. Shaw millisecond trajectories (Shaw et al. 2010) would permit more biochemically meaningful evaluation.

Future work. The immediate targets are (i) scaling to $N_{\text{res}} \in \{150, 200\}$ within mdCATH, (ii) an apples-to-apples comparison with Timewarp and bioEmu on matched tasks, (iii) application to CASP Refinement targets, and (iv) investigating whether deeper GNN architectures or graph-transformer hybrids can close the gap with AlphaDynamics on richer protein contact graphs (beyond the linear chain tested here).

6. Reproducibility

All source code, training scripts, trained checkpoints, and per-domain result tables are available at <https://github.com/krisss0mecom/AlphaDynamics>. Raw MD trajectories are not redistributed; they can be downloaded freely from the mdCATH dataset on Hugging Face (compscience lab/mdCATH). Reproduction of the headline benchmark requires approximately 2 h on a single RTX-5090.

7. Conclusion

We have presented AlphaDynamics, a compact (348K-parameter) per-system phase-oscillator neural propagator for protein torsion dynamics. On a 57-domain subset of the mdCATH benchmark with a strictly uniform simulation protocol, AlphaDynamics outperforms a matched MLP baseline on every domain (mean NLL ratio $6.4\times$ at $N_{\text{res}} = 50$, $7.5\times$ at $N_{\text{res}} = 100$) and

preserves long-rollout structure without explosion, though rollout distributional fidelity remains an open challenge. The architecture—coupled phase oscillators on the torus, evolved via a learnable ODE inspired by the Kuramoto model and phase-gate computing (Gwóźdz 2026a)—demonstrates that torus-native inductive bias yields strong per-system surrogates with minimal parameter budgets. Whether this per-system approach can be extended to transferable models that generalise across protein families is an open and important question for future work.

Figures

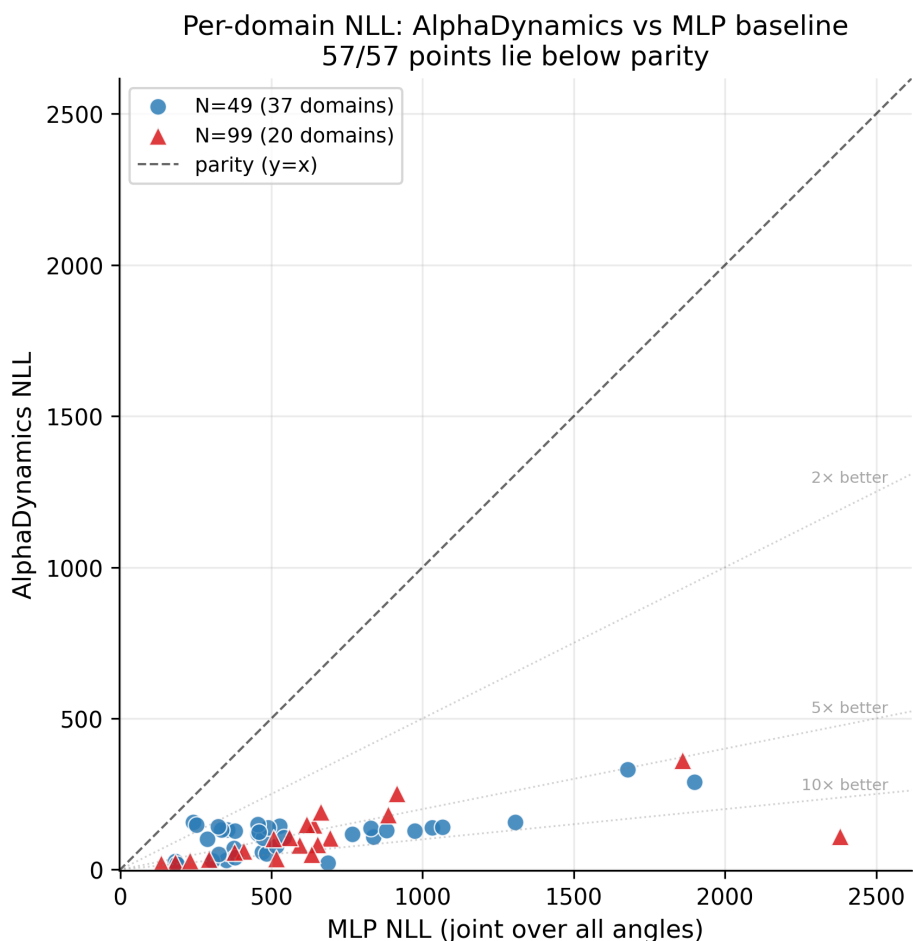


Figure 1 — Per-domain NLL scatter: MLP (x-axis) vs AlphaDynamics (y-axis). Circles: $N=50$ domains; triangles: $N=100$ domains. All 57 points lie below the parity diagonal; most lie below the $5\times$ line.

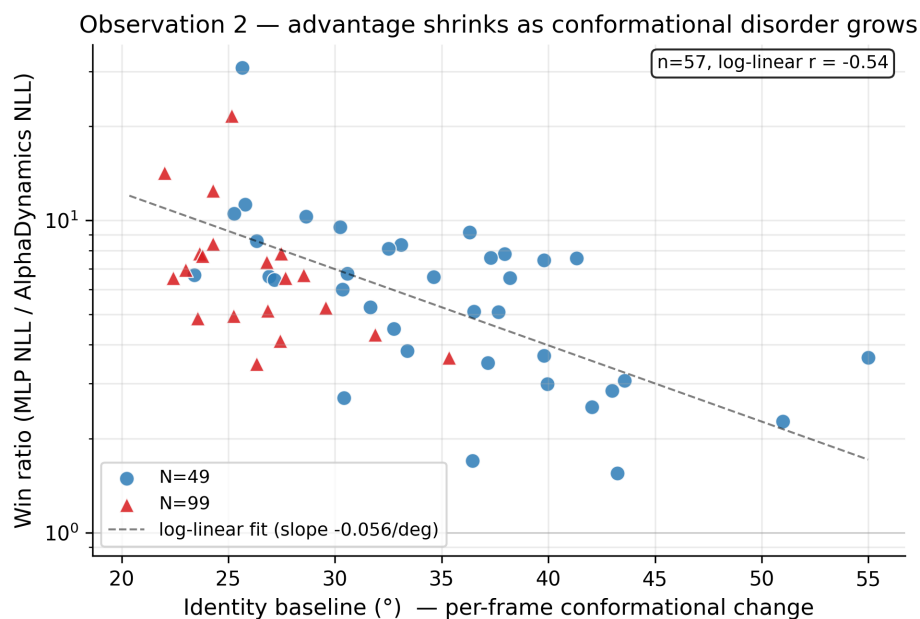


Figure 2 — Observation 2: win ratio $\text{NLL}_{\text{MLP}}/\text{NLL}_{\text{AlphaDynamics}}$ versus per-domain identity baseline. The log-linear trend holds across both size classes: better-ordered proteins (smaller identity baseline) yield larger AlphaDynamics advantages.

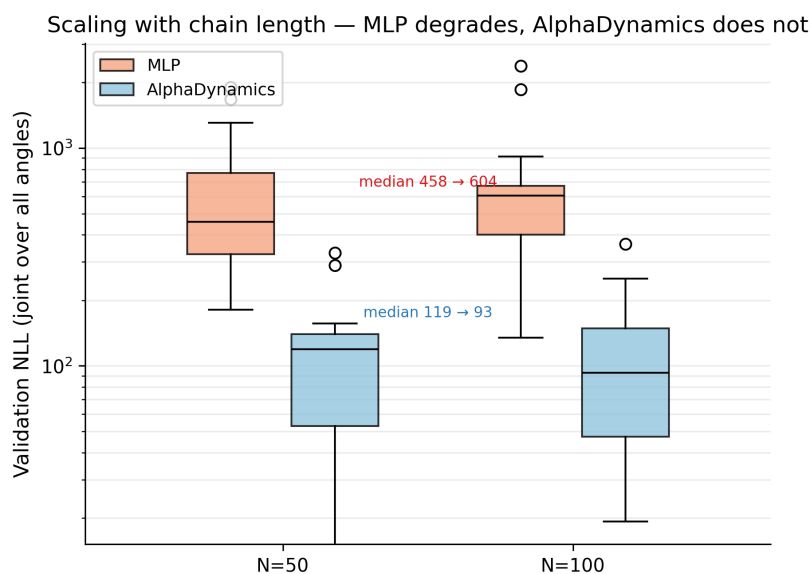


Figure 3 — Scaling behaviour from $N_{\text{res}} = 50$ (37 domains) to $N_{\text{res}} = 100$ (20 domains). MLP NLL distribution shifts upward substantially; AlphaDynamics NLL is nearly unchanged.

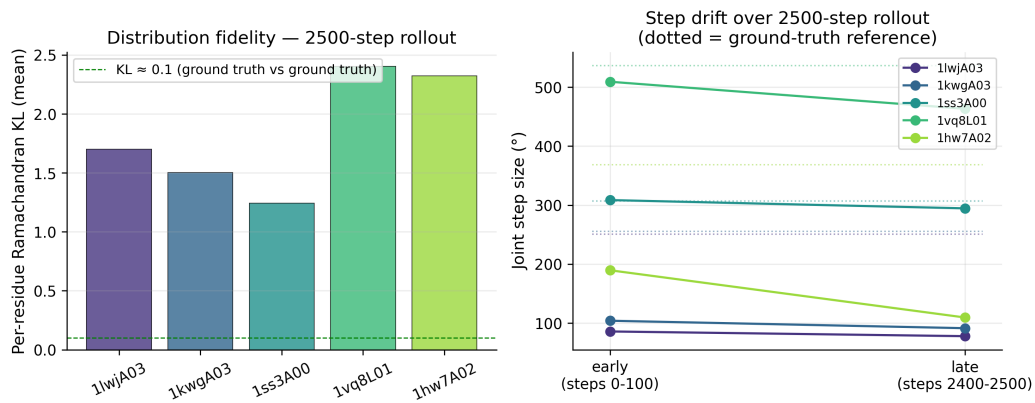


Figure 4 — Long-rollout stability. Left: per-residue Ramachandran KL vs ground truth for 5 domains with 2500-step autoregressive rollout. Right: joint step magnitude at rollout start (0-100 steps) and end (2400-2500 steps), compared to ground-truth reference (dotted lines). Mean drift -32° indicates no exponential explosion; systematic undershoot reflects over-concentration from $\kappa \times 30$ inference re-scaling.

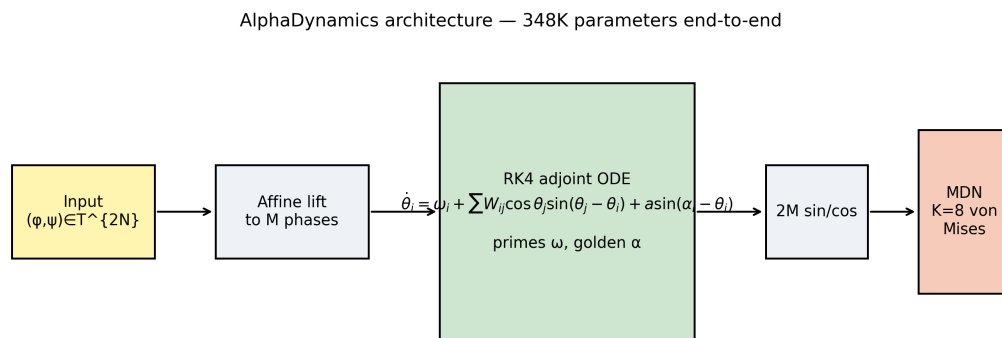


Figure 5 — AlphaDynamics architecture. Input torsions are lifted to M oscillator phases, evolved by a CNOT-coupled phase-flow ODE with learnable frequencies and anchors, and read out as a K-component mixture of axis-independent von Mises densities on the torus.

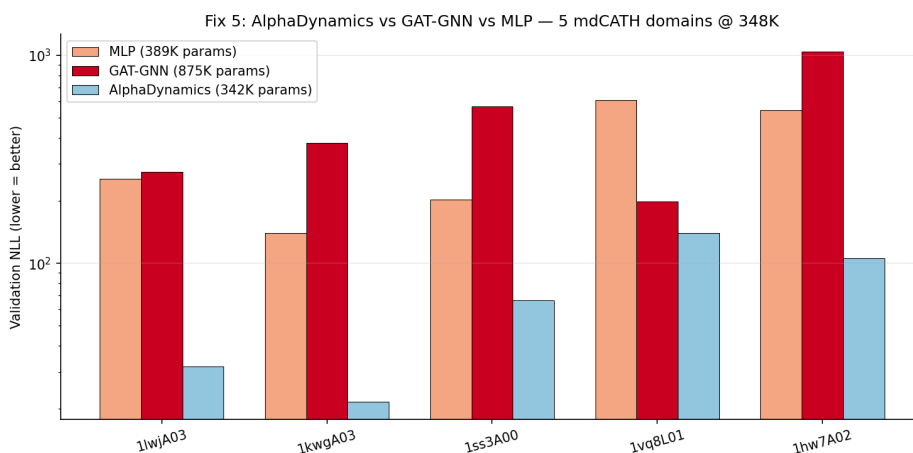


Figure 6 — GNN baseline comparison. Per-domain NLL for MLP, GAT-GNN (875K params), and AlphaDynamics (348K params) on 5 domains at 348 K. AlphaDynamics wins 5/5 against both baselines;

GAT-GNN performs worse than MLP on 4/5 domains despite $2.3\times$ more parameters.

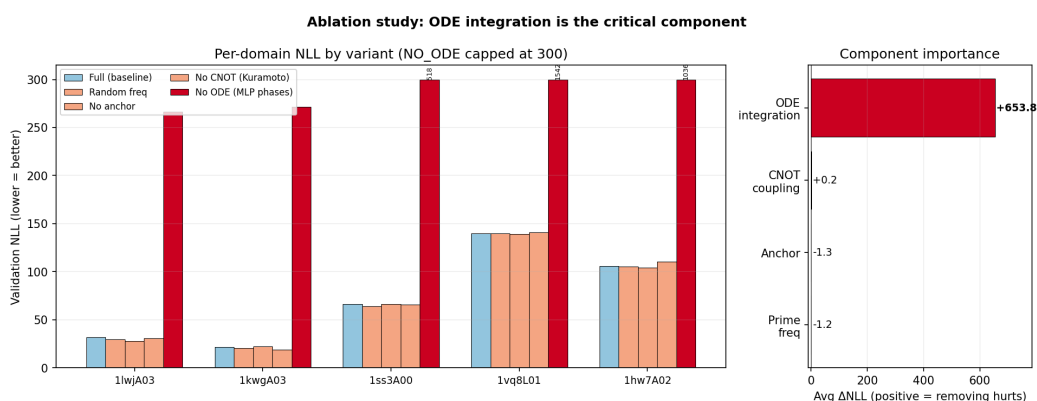


Figure 8 — Component ablation. Removing ODE integration degrades NLL by 654 nats on average ($10\times$). Other components (frequency initialization, anchor, coupling form) have negligible impact.

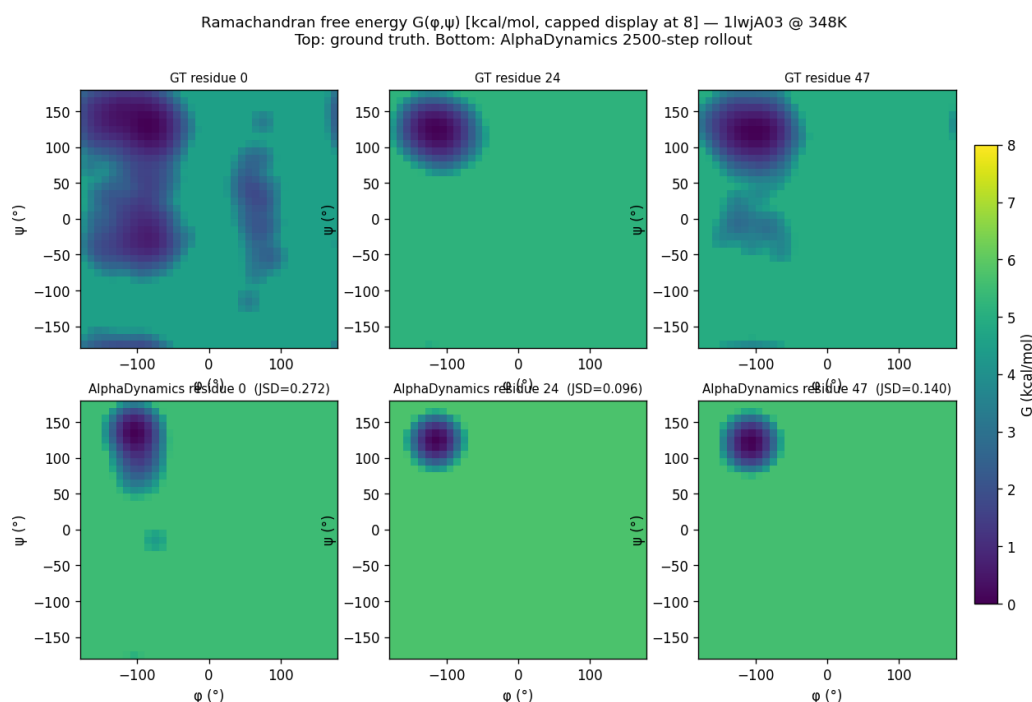


Figure 7 — Ramachandran free-energy maps. Per-residue $G(\phi, \psi)$ from 2500-step AlphaDynamics rollouts (left) vs ground-truth MD (right) for representative residues of 1lwjA03 and 1kwgA03. Basin locations and depths are well reproduced; the main discrepancy is slight over-concentration from the k-rescaling heuristic.

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